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23R-D25 (MAT5404C)

2025

**MATHEMATICS**

(Major-IV)

Paper : MAT5404C

**(Linear Algebra)**

Full Marks : 60

Time : 2½ hours

***The figures in the margin indicate full marks for the questions.***

1. Answer the following questions : 1×7=7

(a) Let  $0_V$  be the additive identity of a vector space  $V(F)$ . Show that  $\alpha 0_V = 0_V$

$\forall \alpha \in F$ .

✓(b) Show that the vectors  $e_1 = (1, 0, 0)$ ,  
 $e_2 = (0, 1, 0)$ ,  $e_3 = (0, 0, 1)$  form a  
spanning set of  $\mathbb{R}^3$ .

(c) What do you mean by a row space of an  $m \times n$  matrix  $A$  over a field  $K$ ?

(d) Check linear dependency of a single non-zero vector  $v$  in a vector space  $V(F)$ .

(e) Find  $p$  so that  $u = (1, 2, p, 3)$  and  $v = (3, p, 7, -5)$  in  $\mathbb{R}^4$  are orthogonal.

(f) Is the vector  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$  an eigenvector of

$A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$  corresponding to the eigenvalue 4?

(g) Define rank and nullity of a linear mapping.

2. Answer the following questions:  $2 \times 4 = 8$

(a) Is the set  $\{(x, y, z) : x, y, z \text{ are rational numbers}\}$  a subspace of  $\mathbb{R}^3(\mathbb{R})$ ?

(b) Let  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the mapping defined by  $T(x, y) = (x + 2, y + 3)$ . Is  $T$  linear? Justify.



✓(c) Check whether the vectors  $u = (1, 1, 1)$ ,  $v = (1, 2, -3)$ ,  $w = (1, -4, 3)$  in  $\mathbb{R}^3$  are orthogonal to each other or not. Are they orthonormal? Give reason.

(d) Define vector space.

3. Answer the following questions :

✓(a) What do you mean by linearly dependent and independent set of vectors in a vector space? Show that the set of vectors  $\{(1, 1, 0), (1, 3, 2), (4, 9, 5)\}$  is linearly dependent in  $\mathbb{R}^3$ . 5

(b) Let  $F : \mathbb{R}^4 \rightarrow \mathbb{R}^3$  be the linear mapping defined by

$$F(a, b, c, d) = (a - b + c + d, 2a - 2b + 3c + 4d, 3a - 3b + 4c + 5d)$$

Find a basis and rank of  $F$ . 5

**Or**

Prove that there exists a basis for each finite dimensional vector space.

✓(c) Using Gram Schmidt orthogonalisation find an orthonormal basis of a subspace  $U$  in  $\mathbb{R}^4$  spanned by the vectors

$$V_1 = (1, 1, 1, 1), V_2 = (1, 1, 2, 4), V_3 = (1, 2, -4, -3)$$

5



**Or**

Find the co-ordinate vector of

$A = \begin{bmatrix} 2 & 3 \\ 4 & -7 \end{bmatrix}$  in the real vector space

$M = M_{22}$ , relative to the basis

$$S = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}$$

4. Solve (a) **or** [(b) and (c)] : 10

(a) Define rank of a matrix  $A$ . Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 2 & 4 & 3 & 7 & 7 & 4 \\ 1 & 2 & 2 & 5 & 5 & 6 \\ 3 & 6 & 6 & 15 & 14 & 15 \end{bmatrix}$$

(i) Find the dimension of row space of  $A$ .

(ii) Find rank  $(M_k)$ ,  $k = 1, 2, \dots, 6$  where  $M_k$  is the submatrix of  $A$  consisting the 1st  $k$ - columns  $C_1, C_2, \dots, C_k$  of  $A$ .



(iii) Which columns  $C_{k+1}$  are linear combination of preceding columns  $C_1, C_2, \dots, C_k$  ?

(iv) Find columns of  $A$  that form a basis for the column space of  $A$ .  
10

(b) Find the characteristic polynomial and minimal polynomial of  $A$  where

$$A = \begin{bmatrix} 4 & -2 & 2 \\ 6 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix} \quad 5$$

(c) Show that any orthogonal set  $S$  of non-zero vectors is linearly independent.  
5

5. Answer **either** [(a) and (b)] **or** [(c) and (d)] :  
10

(a) In an inner product space show that

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2 \quad 5$$

(b) Let  $W$  be the subspace of  $\mathbb{R}^5$  spanned by  $u = (1, 2, 3, -1, 2)$  and  $v = (2, 4, 7, 2, -1)$ . Find a basis of the orthogonal complement  $W^\perp$  of  $W$ .  
5



(c) Consider the following two bases of  $\mathbb{R}^2$ .

$$S = \{u_1, u_2\} = \{(1, 2), (3, 5)\} \text{ and}$$

$$S' = \{v_1, v_2\} = \{(1, -1), (1, -2)\}$$

Find the change-of-basis matrix  $P$  from  $S$  to the basis  $S'$ . 5

(d) Determine whether  $(1, 1, 1, 1)$ ,  $(1, 2, 3, 2)$ ,  $(2, 5, 6, 4)$ ,  $(2, 6, 8, 5)$  form a basis of  $\mathbb{R}^4$ . If not find the dimension of the subspace they generate. 5

6. Answer **either** (a) **or** [(b) and (c)] : 10

(a) What do you mean by eigenvalues and eigenvectors of a linear operator?

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be defined by

$$T(x, y, z) = (2x + y - 2z, 2x + 3y - 4z, x + y - z).$$

Find all eigenvalues of  $T$  and find a basis of each eigenspace. Is  $T$  diagonalizable? If so find basis  $S$  of  $\mathbb{R}^3$  that diagonalize  $T$  and find diagonal representation of  $D$ . 10

(b) Find the dual basis  $\{\phi_1, \phi_2\}$  of the basis  $\{v_1, v_2\} = \{(2, 1), (3, 1)\}$  of  $\mathbb{R}^2$ . 5

- (c) Consider the linear operator  
 $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined by  
 $F(x, y) = (2x + 3y, 4x - 5y)$ .

Find the matrix representation of  $F$   
relative to the basis  $S = \{(1, 2), (2, 5)\}$ .

5

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